Every Minute Counts

An experiential study of Dijkstra's shortest path algorithm

Starting point: JAIN (Deemed-to-be University), School of Commerce, 44/4 District Fund Road, behind Smart Bazaar, Jayanagara 9th Block, Bengaluru 560041.

Study date: 17 October 2026. Road data: OpenStreetMap locations and OSRM car-profile routes. Google Maps: observed recommended driving routes at approximately 22:28-22:30 IST.

Research question

How do human choices compare with optimal routes in a small weighted graph, and how do the answers change when distance, estimated driving time, or corridor availability changes?

Completed numerical findings

Destination	Distance-optimal path	Distance	Time-optimal path	Profile time
C	A-B-C	1.497 km	A-B-C	2.18 min
D	A-B-C-D	2.695 km	A-B-C-D	3.61 min
J	A-K-J	1.847 km	A-B-J	3.04 min
L	A-B-C-L	4.608 km	A-B-C-L	6.14 min

What is complete: 12 locations, 22 selected bidirectional corridors with 44 independent directional weights, two routing objectives, manual traces, local code timing, Google Maps observations, controlled scenarios and classroom materials.

What remains unmeasured: human route choices, decision times, manual completion times, participant familiarity, and any field observations or GPS photographs. No human result or survival benefit has been invented.

Interpretation limits

Optimal means optimal in the selected teaching graph. Edges are pre-routed corridors through real roads, not individual road segments. Their distances follow OSRM profile-selected routes, so this study does not establish the shortest possible distance over every Bengaluru street. Static profile times are not live traffic ETAs or ambulance response times. [S2, S3]

Phase 1 - Locations and road references

The campus is the source in every task. Transit and public destinations provide varied graph structure; the three hospital destinations C, D and L support the fictional campus-to-hospital routing scenario. No assumption is made about clinical suitability or ambulance availability.

ID	Selected location	Mapped latitude, longitude	Road snap
A	JAIN University (source)	12.918126, 77.591874	29.8 m
B	Ragigudda bus stop	12.916799, 77.593385	4.1 m
C	Jayadeva hospital	12.918117, 77.599240	71.5 m
D	Sagar Hospitals	12.927948, 77.599850	41.6 m
E	Jayanagar TTMC	12.928171, 77.583868	2.4 m
F	Jayanagar metro bus stop	12.928771, 77.580345	3.8 m
G	RV Road station reference	12.921988, 77.580046	6.8 m
H	Banashankari bus stop	12.916631, 77.573069	4.1 m
I	Ranga Shankara	12.911608, 77.587080	23.7 m
J	Gopalan Innovation Mall	12.914339, 77.599454	39.8 m
K	Ragigudda Temple	12.914189, 77.593330	67.2 m
L	NIMHANS hospital reference	12.938632, 77.594410	200.5 m

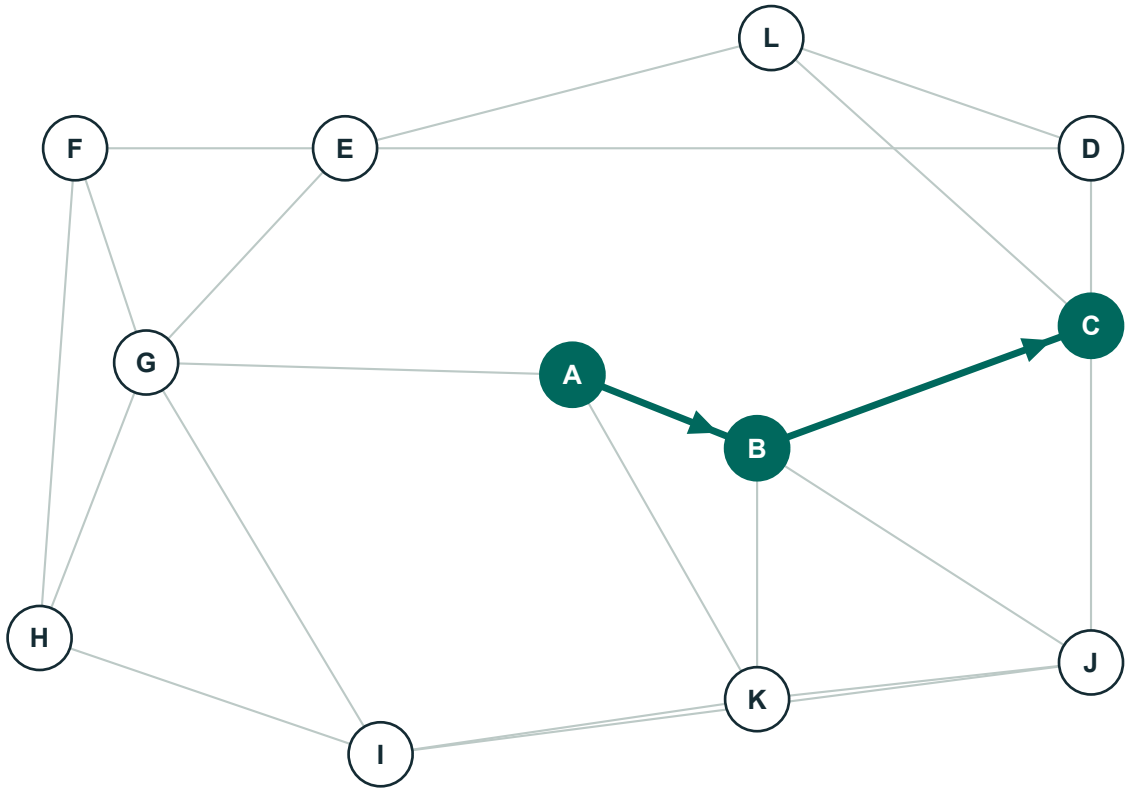
Location handling

Coordinates were retrieved from Nominatim/OpenStreetMap feature records. OSRM snapped them to its car-accessible road network. All route metrics end at those snapped references; the offsets above are not included in route distance or time. Hospital and temple entrances were not verified. L has the largest offset, about 200 m. [S1, S2]

F is a bus stop named Jayanagara Metro Station, not a rail-platform coordinate. G is a station reference snapped to a road. H is the bus stop toward Kadirenahalli. These exact feature types and OpenStreetMap object IDs are preserved in locations.csv.

Google Maps was queried using the snapped road coordinates. B produced an elevated-flyover interpretation: its observed 11.3 km / 22 minute result is retained as evidence of an endpoint ambiguity and excluded from normal cross-service comparison.

Phase 1 - The 12-node teaching graph



Highlighted baseline path: A-B-C. Lines show selected teaching corridors and are not street geometries. Both directions exist, with different costs. Crossing lines do not imply extra graph junctions. [S2]

ID	Location	ID	Location
A	JAIN University (source)	G	RV Road station reference
B	Ragigudda bus stop	H	Banashankari bus stop
C	Jayadeva hospital	I	Ranga Shankara
D	Sagar Hospitals	J	Gopalan Innovation Mall
E	Jayanagar TTMC	K	Ragigudda Temple
F	Jayanagar metro bus stop	L	NIMHANS hospital reference

Model choices

The teaching graph is a sparse selection of 22 landmark-to-landmark connections. Each edge uses a cell from the 12 x 12 OSRM table. Roads, intermediate junctions and legal restrictions are handled inside that pre-routed corridor; the teaching graph exposes only its endpoints and cost.

Separately routed corridors can overlap physically. Concatenating them omits turn-history restrictions at landmark boundaries and can introduce unnecessary stops or reversals. This makes the graph suitable for algorithm learning, while a production road network would require junction-level modeling and explicit access validation.

Phase 1 - Complete corridor metrics

Forward means the first letter to the second letter. Reverse means the second letter to the first. Distances are metres; durations are seconds from a static car-routing profile. These are road-route estimates rather than surveyed distances or observed travel times.

Corridor	Forward m	Reverse m	Forward s	Reverse s
AB	618.5	250.7	58.7	28.8
AK	829.9	821.7	83.3	83.9
AG	2053.7	2358.6	171.2	179.8
BC	878.5	1400.0	72.2	121.6
BK	624.2	571.0	59.1	55.1
BJ	1367.9	953.2	123.7	71.1
CD	1198.0	2107.0	85.8	202.2
CJ	1434.1	1067.1	118.0	95.0
CL	3110.9	3370.4	237.3	307.5
DE	2435.5	2368.2	234.8	221.1
DL	1996.7	2072.5	174.2	195.0
EF	771.2	454.3	68.3	51.6
EG	1571.5	1131.8	150.4	109.8
EL	2611.6	2306.0	229.4	236.0
FG	1228.6	817.5	119.4	71.9
FH	1974.3	1974.9	146.9	182.7
GH	1523.2	1511.9	120.5	172.6
GI	2196.0	1850.1	173.8	174.4
HI	2657.1	2137.0	245.0	160.3
IK	1227.2	1648.3	141.1	157.1
IJ	1959.0	2465.0	164.6	189.7
JK	1164.6	1017.4	103.8	108.6

Source: OSRM table API, annotations=distance,duration. Full request, 12 x 12 matrices, returned snapped coordinates and raw response are included in data/. Reverse costs are not copied from forward costs. [S2, S3]

Phase 2 - Dijkstra trace and correctness

Manual example: minimize distance from A (JAIN) to C (Jayadeva hospital road reference). Initialize A=0, all other costs=infinity. Each iteration settles the smallest tentative cost, then relaxes outgoing edges. Stop when C is settled.

Step	Settle	Cost (m)	Improved tentative labels
1	A	0.0	B=618.5 via A, G=2053.7 via A, K=829.9 via A
2	B	618.5	C=1497.0 via B, J=1986.4 via B
3	K	829.9	I=2478.2 via K, J=1847.3 via K
4	C	1497.0	No updates; target settled

Path reconstruction

Following predecessors from C back to A gives C <- B <- A. Reverse this sequence to obtain A-B-C. Its cost is 618.5 + 878.5 = 1,497.0 metres. The same path has a static profile duration of 130.9 seconds, or 2.18 minutes.

Implementation

The Python implementation uses a min-heap, tentative costs, predecessor pointers, stale-entry checks and a settled set. A deterministic node-ID tie break makes traces reproducible. With a binary heap, the usual bound is O((V+E) log V); this tiny graph is for understanding correctness, not demonstrating city-scale performance.

Verification completed

An independent Floyd-Warshall calculation checked all 144 origin-destination pairs for distance and again for duration: 288 comparisons. Path costs were reconstructed and checked. Additional cases cover an unreachable destination, source equal to destination, zero weight, equal-cost alternatives and negative-weight rejection.

NetworkX is optional in build_study.py. It was not installed in the available runtime, so no NetworkX verification is claimed. When installed, the script additionally checks the source-to-destination costs against NetworkX.

Timing method

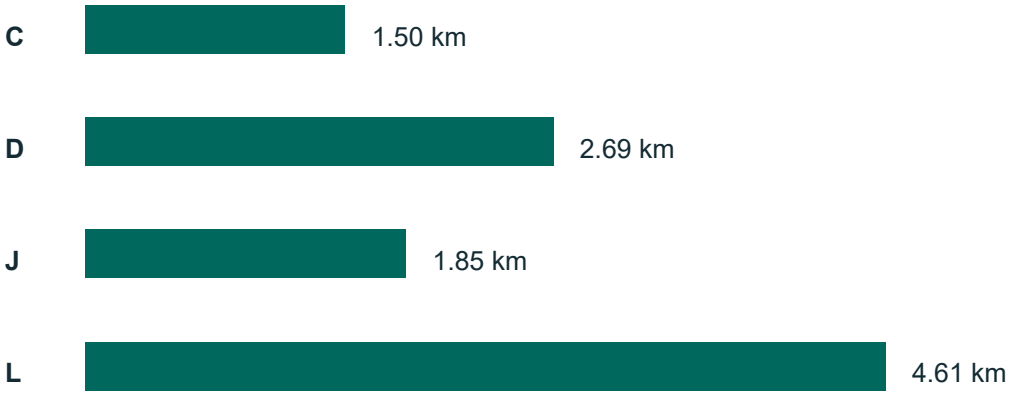
Each destination/objective was measured using 21 batches of 100 calls. results.csv records the median and nearest-rank 95th percentile of the per-batch mean milliseconds. The graph is prebuilt; file access, network calls, human work and trace recording are excluded. Values describe this local Python environment only.

Phase 2 - All destination comparisons

Teaching results are optimal in the selected graph. OSRM direct means a route over the larger underlying road network using the same road references. Google values are its first recommended driving route, using Leave now; distances and ETAs are rounded as displayed.

ID	Graph km	Graph min	OSRM km	OSRM min	Google km	Google min
B	0.619	0.98	0.619	0.98	11.3*	22
C	1.497	2.18	1.084	1.61	1.1	5
D	2.695	3.61	2.282	3.04	2.1	7
E	3.186	4.68	2.523	3.30	2.5	8
F	2.871	4.05	2.824	3.35	2.9	8
G	2.054	2.85	2.054	2.85	2.1	7
H	3.577	4.86	2.341	2.62	2.4	8
I	2.478	4.01	1.588	2.29	1.6	6
J	1.847	3.04	1.574	2.47	1.6	6
K	0.830	1.39	0.830	1.39	0.7	3
L	4.608	6.14	2.496	4.18	2.6	9

Graph km uses the distance-optimal route. Graph min uses the time-optimal route; they can refer to different routes. * B is excluded from ordinary comparison because of the surface-road/flyover ambiguity.



Graph distance for core tasks C, D, L and the additional objective-comparison task J. Human results are intentionally absent.

What the differences mean

For L, the teaching graph takes 4.608 km, compared with the OSRM direct reference of 2.496 km. This exposes the cost of the chosen corridor set. It does not indicate an error in Dijkstra. Google additionally uses a different road model and ETA assumptions, so neither its distance nor its time is a ground-truth check of this algorithm.

Phase 2 - Objective and disruption experiments

Shortest versus fastest: destination J

Objective	Route	Distance	Profile duration
Distance	A-K-J	1847.3 m	191.9 s
Duration	A-B-J	1986.4 m	182.4 s

This example changes the route when the optimization objective changes. A-B-J is longer than A-K-J but has a smaller modeled duration. The two measurements must therefore be reported separately.

Controlled BC-corridor scenarios

Target	Scenario	Time-optimal route	Profile min
C	baseline	A-B-C	2.18
C	BC corridor unavailable	A-B-J-C	4.62
C	BC duration tripled	A-B-C	4.59
D	baseline	A-B-C-D	3.61
D	BC corridor unavailable	A-B-J-C-D	6.05
D	BC duration tripled	A-B-C-D	6.02
L	baseline	A-B-C-L	6.14
L	BC corridor unavailable	A-G-E-L	8.51
L	BC duration tripled	A-G-E-L	8.51

Scenario interpretation

Unavailable BC removes both B-to-C and C-to-B teaching arcs. The delay experiment multiplies the duration of those arcs by three and leaves their distance unchanged. Each scenario uses fixed weights during one run. These are designed sensitivity experiments, not observed roadworks or measured traffic conditions.

A physical road closure could also affect other overlapping corridors. The present scenario does not reroute the underlying road network. For a street-level closure study, remove the affected road segments and recalculate all relevant corridor routes.

Phase 3 - Emergency-service reality check

Historical Karnataka context

A Times of India report published on 9 December 2020 summarizes a CAG audit covering 2014-15 to 2018-19. It reports an intended mean response of 20 minutes in urban areas and 30 minutes in rural areas, and says 72% of cases met the audit's 30-minute threshold. The report also states that rural/urban classifications were unavailable for that analysis. This is a secondary account of a historical audit, not a current Karnataka service standard or performance estimate. [S4]

Figure	Meaning	Permitted use
20 min / 30 min	Historical urban/rural mean targets as reported in 2020	Explain why metric definitions and reporting periods matter
72% within 30 min	Historical proportion reported for audited cases	Discuss aggregate performance; do not compare directly with one simulated route

A documented example of metric definitions

The NHS England AmbSYS specification dated 25 July 2022 defines call connection, call answer and response clock rules. It lists separate category-specific mean and percentile standards, such as C1 mean 7 minutes and 90th percentile 15 minutes. These are historical English system definitions, included to illustrate measurement discipline; they are not Bengaluru benchmarks. [S5]

Why route time is only one component

A route model covers travel between selected road points. Operational response can additionally depend on call handling, triage, vehicle availability, dispatch and departure. Clinical care, scene work and hospital handover introduce further intervals. A short graph-route estimate cannot establish a short complete response.

Prepared reality-check conclusion

Our graph demonstrates optimal routing under declared assumptions. Historical service reports show that ambulance performance is assessed across many incidents and includes operational processes that our graph omits. We therefore use these reports to identify missing variables, rather than to claim that the classroom algorithm improves emergency outcomes. Human-versus-algorithm conclusions remain pending until participant observations are collected.

Phase 4 - Session and presentation guide

Minutes	Activity	Evidence
0-15	Introduce campus routing, source A, objectives and simulated scenario	Written predictions
15-45	Build the directed network and inspect coordinate/road snapping	Annotated graph and sourced edge table
45-65	Collect independent human guesses before revealing answers	Chosen paths, decision times and familiarity
65-95	Manually trace A to C; reconstruct predecessors	Completed manual trace and timed completion
95-120	Run Python; compare with the independent check; inspect task J	Route results, code timings and objective comparison
120-140	Disable BC or triple its duration; predict and rerun	Before/after scenario table
140-160	Review Google observations and historical service metrics	One-paragraph reality check
160-180	Present findings and reflect on limitations	Short presentations and exit responses

Human experiment protocol

Use at least several participants across C, D and L; J is an optional extension. Everyone sees the same baseline graph, weights and instructions. Do not reveal algorithm results until guesses are recorded. Record task order and prior area familiarity. Use fresh participants or counterbalanced task order to reduce learning effects. Record a timed-out or invalid route explicitly rather than inventing a cost.

Measures and analysis

Keep route distance (m), modeled route duration (s), human decision time (s), manual completion time (s), and code computation time (ms) in separate columns. Route gap = 100 x (chosen cost - optimal cost) / optimal cost, with the same objective in numerator and denominator. An optimal human route has 0% gap and counts as success. When data arrive, report participant count, median gap and the proportion choosing an optimal route; avoid causal or population-wide claims from a small convenience sample.

Presentation sequence

The supplied presentation covers the question, study network, model assumptions, manual trace, completed route results, objective comparison, BC scenarios, Google Maps context, emergency-service metrics, and the pending human experiment. Add real participant findings after collection. Assign mapping/data, algorithm, experiment/analysis and communication responsibilities, with every member completing a trace.

Printable student worksheet

Participant code: _____ Task: C / D / L / J Objective: distance / duration

Before solutions are shown, write a route from A: _____

Decision time: _____ seconds Familiarity with area: _____

Manual trace completion time: _____ seconds Task order: _____

Trace table: use metres for distance or seconds for duration. Initialize A=0; all others=infinity. Fill predecessor changes in a separate column or beside the graph.

Step	Settled	Tentative updates and predecessors

Reconstructed path: _____

Chosen route cost: _____ Optimal cost: _____ Route gap: _____

Reflection

1. Which edge changed your initial choice, if any?
2. Why might the shortest and fastest routes differ?
3. Which unmodeled factor could change an actual trip?
4. Did the Google Maps endpoint and road level match the intended point?

Field evidence, if collected later

Use public-junction photographs with timestamp and location tags to document observations. Record feature ID, capture time, road level, visible access restriction and image filename. These photographs are not travel-time measurements. No field photographs are claimed in this report.

Sources, reproducibility and submission checklist

S1 - JAIN official campus location page

https://www.jainuniversity.ac.in/contact/top-indian-university-with-best-colleges-with-location-maps

Confirms #44/4 District Fund Road, Jayanagar 9th Block, Bengaluru 560041.

S2 - OpenStreetMap / Nominatim and OSRM responses

https://www.openstreetmap.org/copyright

Each exact feature URL and query is in locations.csv. Raw geocodes and the complete OSRM matrix are saved; attribution: OpenStreetMap contributors, ODbL 1.0.

S3 - OSRM HTTP API definitions

https://project-osrm.org/docs/v26.6.1/http

Table durations are seconds. Distances are metres along profile-selected routes, not guaranteed shortest street distances. Server build, OSM extract date and profile configuration were not supplied by the public endpoint.

S4 - Times of India, 9 December 2020

https://timesofindia.indiatimes.com/city/bengaluru/karnataka-108-ambulance-service-failed-to-provide-service-within-golden-hour-in-50-trauma-cases-says-cag/articleshow/79648476.cms

Secondary reporting of CAG findings for 2014-15 to 2018-19; primary audit document not independently retrieved.

S5 - NHS England AmbSYS specification, 25 July 2022

https://www.england.nhs.uk/statistics/wp-content/uploads/sites/2/2022/07/20220725-AmbSYS-specification.pdf

Historical definitions and standards. Included for metric interpretation only.

S6 - Google Maps route observations, 17 October 2026

https://www.google.com/maps

Individual URLs, UTC timestamps and notes are in google_maps.csv. Recommended route values are rounded snapshots and can change.

Reproduce and extend

Open index.html directly for the offline explorer. Run python build_study.py to regenerate numerical outputs without a network connection. The data are frozen snapshots, not a live API integration. Add observations to data/human_input.csv, then rerun; valid paths are scored against the same baseline. Build scripts regenerate publication files when the documented artifact dependencies are available.

Submission status

Problem and relevance, graph modeling, implementation, recorded computational findings, visual report, source notes, session materials and a presentation are supplied. The human-results analysis and participant reflection remain pending. They must be added before presenting this as a completed human-versus-algorithm experiment.